

AEMaaCS Admin & DevOps Best Practices

Admins, SREs, DevOps – SHRSS Volume

1. Role & Responsibilities

As an AEMaaCS admin/DevOps engineer for SHRSS, you:

- Own the **health and reliability** of AEM environments.
- Manage **Cloud Manager, pipelines, environments, access, and logging**.
- Serve as the **first line of triage** for production issues (performance, availability, security).
- Enforce guardrails so engineering and authoring teams operate safely.

Use the **Consolidated All-Roles Volume** for shared concepts.

2. Environment & Program Management

2.1 Programs, Environments, and RDEs

- Keep environments clearly labeled and documented:
 - **DEV** – daily integration testing.
 - **STAGE** – pre-production validation.
 - **PROD** – live traffic.
 - **RDE** – rapid experimentation and troubleshooting.
- Use **consistent run modes** and avoid environment-specific hacks; all differences should be in configs.

References

- [Manage environments](#)
- [Rapid Development Environments](#)

SHRSS Overlay

- SHRSS relies heavily on **external integrations** (Workday, DPLT, translation). For each environment:
 - Maintain clear mapping of endpoints and credentials (e.g., Workday sandbox vs production).
 - Use environment variables and OSGi configs; never let devs embed URLs in code.
- Document:
 - **Which integrations are active in which envs** (some may be disabled in DEV).
 - Any **data anonymization** or subset rules in non-prod.

3. Cloud Manager Pipelines

3.1 Pipeline Types & Usage

- **Non-production pipeline**
 - Builds and deploys to DEV.
 - Runs code quality, security, functional tests.
- **Production pipeline**
 - Deploys to STAGE then PROD.
 - Includes performance and experience audits.
 - Requires approvals and change management alignment.

References

- [CI/CD Pipelines](#)

3.2 Quality Gates as Policy

- Treat quality gates as **policy**, not suggestions:
 - Code quality: no Critical or Major issues without documented exceptions.
 - Security: no open critical vulnerabilities.
 - Performance: monitor regressions.

SHRSS Overlay

- Given SHRSS's heavy reliance on **Jobs/Events search**:
 - Monitor code quality findings related to **index definitions** and **query patterns**.
 - If a pipeline fails on custom index rules, escalate to engineering rather than bypassing.
- Ensure **integration tests** in `it.tests` validate:
 - Jobs listing behavior for basic filters.
 - At least one happy-path flow per integration (Workday sync, DPLT sync) where possible.

4. IAM, Access & Permissions

4.1 Product Profiles & AEM Groups

- Use **Adobe Admin Console** for identity and group management:
 - Product profiles: AEM Users, AEM Administrators.
- Map identity groups to AEM groups for permissions:
 - E.g., shrss-authors → content-authors-shrss group in AEM.

References

- [Assigning AEM Product Profiles](#)

SHRSS Overlay

- Codify a **role mapping table**:
 - Marketing authors, Recruiters, DAM librarians, Admins, etc.
 - Map each to AEM groups and product profiles.
- For Jobs & Events:
 - Restrict who can publish to **production-critical CF folders** (e.g., Jobs/Events CFs).
 - Implement review/approval workflows where legally required (e.g., job postings).

5. Dispatcher & Security

5.1 Dispatcher Configuration

- Store all dispatcher configs under the dispatcher module; manage via pipelines.
- Key files:
 - `conf.d/enabled_vhosts` – virtual hosts.
 - `conf.dispatcher.d/enabled_farms` – farms, cache rules, filters.

References

- [Dispatcher Configurations in Adobe Experience Manager as a Cloud Service](#)
- [Dispatcher Overview](#)

5.2 Security Filters

- **Block by default**, then explicitly allow:
 - Block `/system/console`, `/crx/*`, `/bin/*` except whitelisted endpoints.
 - Block author/publish admin endpoints from public internet.
- Ensure HTTPS and HSTS are enabled as required by policy.

SHRSS Overlay

- SHRSS has custom endpoints for:
 - Jobs/Events/News APIs, GraphQL persisted queries, integration callbacks.
- For each endpoint:
 - Confirm it is **explicitly allowed** and **properly cached or non-cached** in dispatcher.
 - Validate:
 - No author-only endpoints are exposed.
 - No Workday/DPLT webhook endpoints are reachable without appropriate access control if applicable.

6. Monitoring, Logs & Troubleshooting

6.1 Logging Strategy

- Use Cloud Manager to:
 - Download logs for historical investigation.
 - Set up **log ingestion** into central observability solutions if available.
- Use Developer Console for:
 - Live log tail.
 - Health checks, status dumps.

References

- [Debugging AEM as a Cloud Service with the Developer Console](#)

6.2 Performance & Query Tools

- Use:
 - **Query Performance Tool / Query Analyzer** for slow queries.
 - **Oak logs** and warning messages such as “query without index detected.”
- Coordinate with engineering to:
 - Add or refine index definitions.
 - Simplify or constrain high-cost queries.

SHRSS Overlay

- Periodically examine top queries on **Jobs**, **Events**, and **News**:
 - Look for scans over /content/dam without path constraints.
 - Look for fulltext queries without appropriate indexes.
- When Jobs search performance regresses:
 - First check queries via Query Analyzer.
 - Then ensure the damAssetLucene-* customizations deployed for SHRSS are in place and not reversioned.

7. Change & Release Management

- Tie every deployment to a **change record** with:
 - Description, risk, rollback strategy.
- Maintain a **runbook** for:
 - Common incidents (slow Jobs search, missing Events, dispatcher cache issues).
 - Rolling back to previous artifacts and invalidating caches.

SHRSS Overlay

- Ensure runbooks include:
 - Paths to critical CF folders (Jobs, Events, News, Locations).
 - Index names and locations for SHRSS customizations.
 - Known troubleshooting flows (e.g., how to verify Workday sync status and logs).

8. Admin & DevOps SHRSS Overlay Summary

- Treat SHRSS as a **search-heavy, integration-heavy AEMaaCS solution**, not just basic WCM.
- Invest in **pipeline quality, dispatcher security, and index monitoring** as first-class ops concerns.
- Keep **clear documentation** of environment differences, integration endpoints, and runbooks.

Use this alongside:

- The **Consolidated All-Roles Volume**.
- The **Engineering Volume** (for code-level practices).
- The **Indexing & Performance Volume** (for query/index ops details).